

COMMUNICATION CODE

Development adequate contextual reactions (ACR) in LLM



Deficit groups in ASD

Social manifestations and perceptions of neurotypical people

Social and communicative deficits

- Reduced empathy
- Echolalia
- Monotony of speech

Behavioral features

- Rituals and stereotypes
- Rigid attachment to things and routine

❏ These manifestations are not personality traits, but rather a consequence of the neurobiological characteristics of information processing. Understanding their nature is the key to effective pedagogical intervention.

True Manifestation: Neural Level

Why do shortages occur?

Sensory entanglement

Distortion of signal perception - the brain misinterprets incoming data

Sensory overload

Excess of incoming information - the system cannot cope with the volume of stimuli

Sensory hunger

Lack of certain stimuli - a deficiency of the necessary signals for normal functioning

Two ways to reduce deficits in theory

1

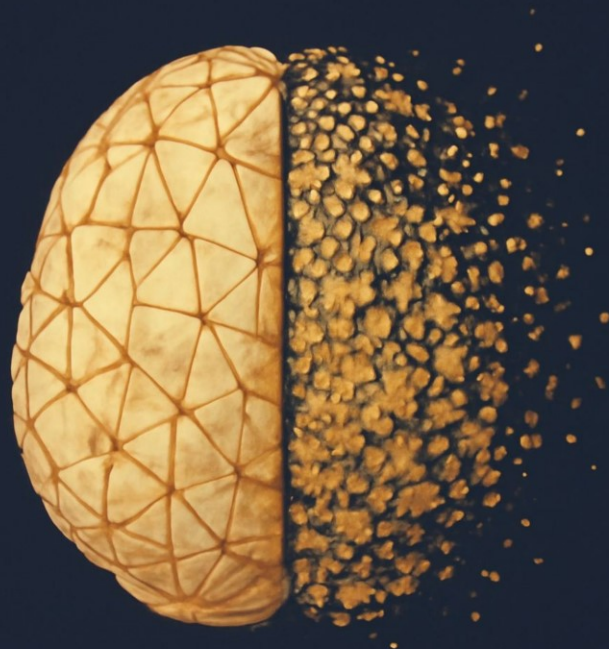
The Inner Path

Neurobiological - impact on signal processing mechanisms

2

External path

Neuropsychological/behavioral – development of compensatory strategies



The problem of reduced empathy

What is the basis?

Appropriate Contextual Response (ACR)

Warp empathy . ACR is ability automatically read context and respond in accordance with situations .

What is context?

Semantic environment events that determine their essence and motive .
Without readings context in a social situation adequate is impossible reaction .

The cause of ACR failure in people with ASD

- The brain does not automatically read social markers.
- Information is processed **from the bottom up** ; focusing on details prevents the collection of a unified context.

- ❑ The neurotypical brain constructs context from the top down—from the general to the specific. In ASD, this process is inverted: details absorb all attention, and the holistic picture fails to emerge.

Traditional methods of solving the problem of ACR

1

Visual support

The TEACCH concept is structuring the environment through visual cues.

2

Social Stories

Social Stories – describing hidden context through short narratives

3

Applied behavior analysis

ABA therapy is functional analysis behavior , partitioning actions on steps with reinforcement

4

Social skills training

Social Skills Groups – practice micro-skills in group games

5

Development of a model of the psyche

Theory of Mind Training — step-by-step algorithms for understanding emotions

Limitations of the TEACCH and Social Stories approaches

Method	The essence	The main disadvantages for the ACR
TEACCH	Replacing intuition with rigid visual logic	Rigidity, dependence on prompts, lack of spontaneity
Social Stories	Describing hidden context through short stories	Mechanical connections, robotic behavior, poor generalization of skills

 Both methods create dependence on external cues and do not develop the ability to independently read the context in new situations.

Limitations of ABA, Groups, and Theory of Mind approaches

ABA therapy: Analysis, breaking down actions into steps

Cons: Training is based on templates, lacking spontaneity. Behavior is reproduced in familiar conditions—there is a problem with skill transfer when changing contexts.

Social Skills Groups: Microskills Practice Through Games

Cons: Rigid scenarios, "laboratory conditions" effect, social camouflage. Participants imitate behavior without mastering the mechanism.

Theory of Mind Training: Step-by-step algorithms for emotions

Cons: Overgeneralization, inability to grasp hidden context. The algorithm fails when the situation goes beyond the accepted framework.

General deficiency of most methods

Violation of the logic of ontogenesis

Problem

The training is built **top-down** —from effect to cause. The methods address symptoms without addressing the neurobiological root of the ACR deficit.

Investigation

Reinforcing rigidity and developing a “**mechanical**” **intelligence** —a set of accepted responses without adaptation to the specifics of the context.

Why can't we go from effect to cause?

This requires a set of template situations that form a **relevant sample**. However, preparing one is impossible for two reasons:

- The uniqueness of each case of ASD (**Weber's principle**)
- The infinite variety of real social contexts

 Weber's principle: individual perception thresholds in people with ASD are so different that a universal template set of situations is fundamentally unattainable.

How to reverse the approach?

Bottom-Up Logic

1

Method of replacement ontogenesis (MZO)

Reproduces stages of normal development. **Limitation:** For ASD, it does not address the problem of ACR, as the root cause lies in sensory confusion, not in missed developmental milestones.

2

Social Thinking

Rational learning through systems analysis and the mathematical logic of cause-and-effect relationships. **A way out of the impasse of traditional methods.**

- ❏ Social Thinking does not bypass sensory entanglement; it offers a rational tool for compensating for it: transforming the social situation into an analyzable system.

Principles of Social Thinking

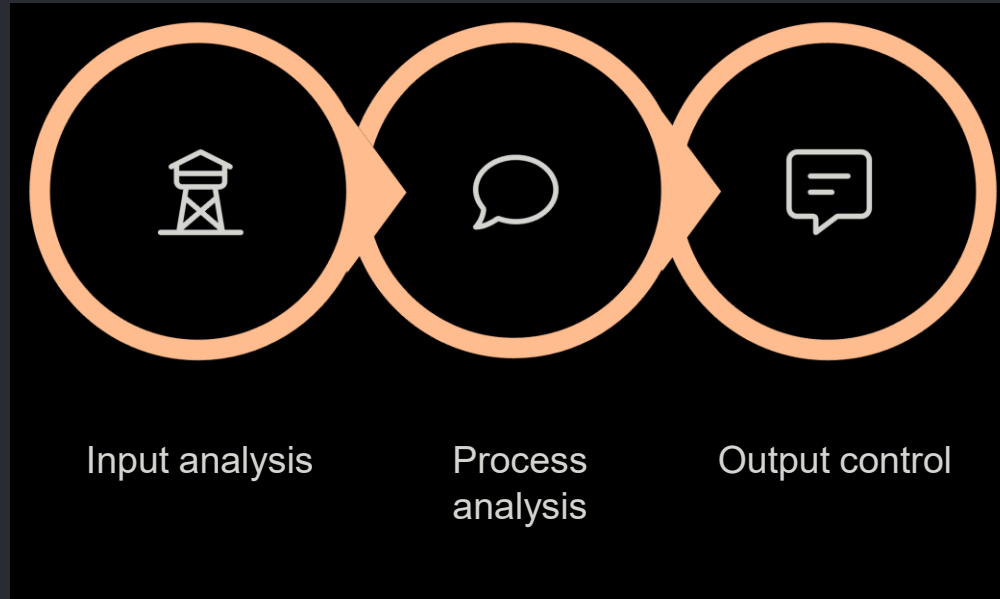
The ILAUGH Model: Social Situation as a Mathematical Problem



The ILAUGH model moves social interaction from the realm of intuition to the realm of analyzable systems—this is what makes it applicable to ASD, where intuitive reading of context is impaired.



General Model of Learning in Social Thinking



The model transforms each social situation into a structured task with input data, processing rules, and a predictable output—this is how the LLM approach works in the "Communication Code" course.

1. Input analysis

Where am I? Who are the elements of the system? What is their state?

2. Process analysis

What is the purpose of the system? What are the rules?
How do others feel?

3. Output control

Predicting system response and feedback loop

LLM as a tool for solving old problems

Why are large language models (AI) changing everything?

High imitation of speech

Achieved through Transformer architecture and deep learning

Solving the camouflage problem

AI allows us to avoid falling into rigid patterns

Solving the Generalization Problem

Provides flexible mastery of the method in a changing virtual environment



Course "CODE OF COMMUNICA

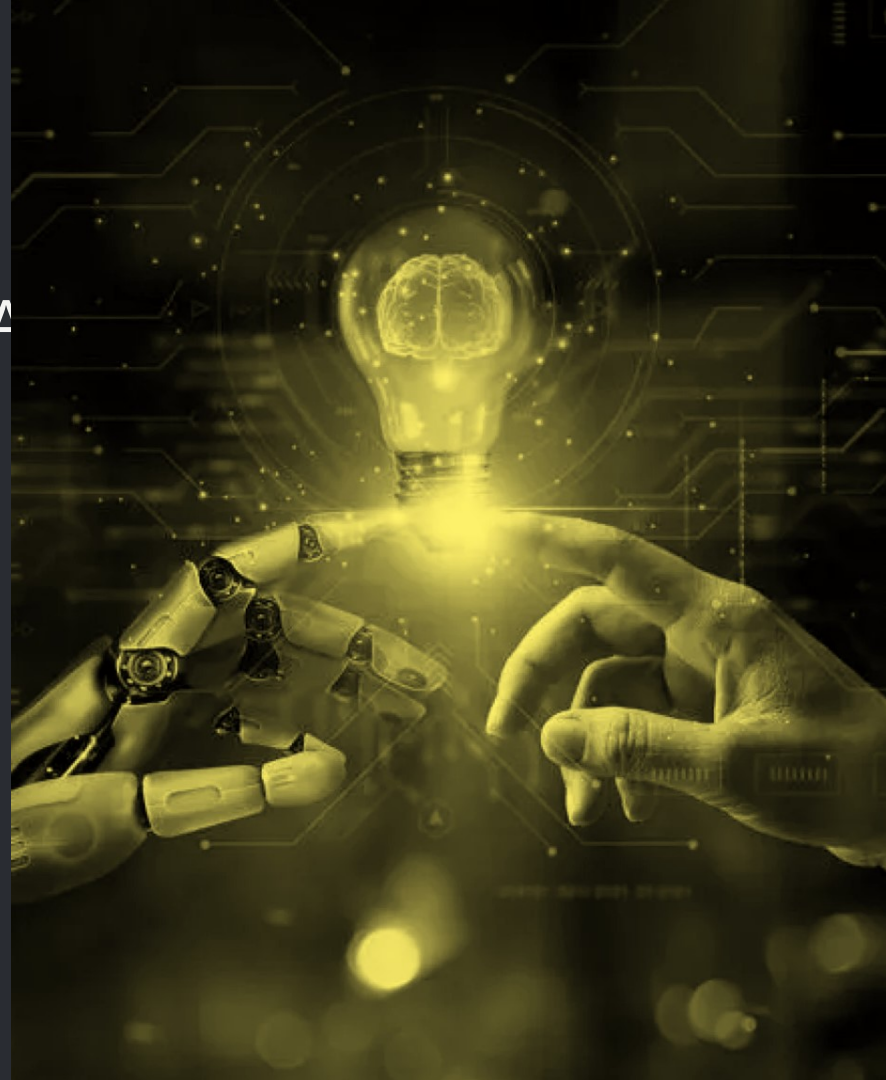
Practical embodiment method

The essence

Education social behavior through Social Thinking approach in virtual LLM

Target

Formation of the ACR through mastery method constructions models communications with AI



Course stages and modules

1

Module 1

Getting to Know Your AI Interlocutor (Anxiety Reduction, The Matrix Pattern)

2

Module 2

Basic social scripts (Hard algorithms)

3

Module 3

Initiative and Dialogue (Development of Flexibility)

4

Module 4

Dealing with difficult situations and refusals

5

Module 5

Final Integration (Mastering the Modeling Method)

A systematic approach to learning

Tools for building a personalized route

Prompting and vectorization

Structuring input data for AI

Component analysis and discretization

Breaking down complex situations into their elements

Matrix method and Euler circles

Visualization of logical connections

Hidden Markov Models

Predicting action sequences

Course Result: Smooth transfer of skills

Key message: "Everything has its own code..."

Why does this work?

The high convergence of AI assistants' actions with the behavior of real people ensures a "smooth transfer" of skills from the virtual environment to life.

The main result

A person with ASD masters a universal method of analyzing social situations that works in any context—from virtual dialogue to real-life interaction.

